

Who's in Charge of Charging Experience: Heuristic Evaluation Criteria for Addressing Design Issues in Electric Vehicle Charging

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ABSTRACT

As the population of electric vehicles (EVs) continues to grow, managing and enhancing the Electric Vehicle Charging Experience (eCX) has become an inevitable challenge. However, the research community has not given comprehensive attention to the eCX, often neglecting interactions between key components: EVs, chargers, and mobile apps. This research addresses the gap by identifying heuristic evaluation criteria to measure the problems of existing EV chargers for effective management. The charging process was analyzed hierarchically to establish criteria, and its relevance to previously established heuristics was evaluated. Domain experts then verified 27 criteria in 7 tasks, whether these can effectively identify issues affecting eCX through ratings and discussions. As a result of quantitative analysis of the ratings and qualitative examination of the discussions, formulated criteria can offer comprehensive and valuable insights spanning various eCX components and tasks. This research can guide designers in enhancing the eCX for current and prospective users.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **HCI design and evaluation methods**; • **Usability testing**; • **Interaction design**; • **Interaction design process and methods**; • **User centered design**;

KEYWORDS

Charging experience of electric vehicles(eCX), Heuristic evaluation criteria, Design guidelines for eCX, Integrated service interaction

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1 INTRODUCTION

In response to the critical issue of global warming [20], the emergence of eco-friendly policies has driven market growth in environmentally friendly vehicles [30, 33]. To comply with these policies, an increasing number of countries are adopting eco-friendly vehicles, including electric vehicles (EVs) and plug-in hybrids, with EVs sales expected to reach 35% by 2030 [9].

Aligned with these policies, research on EVs charging has also increased and is currently centered on enhancing performance quality to pioneer the new market [24]. According to the Kano model [11], the present attractive attribute in EV charging is the satisfaction of user experience (UX), while the must-be attribute is the charger's performance, determining whether it functions properly. Nowadays, even if the UX is not satisfactory, it is still a compelling quality that encourages usage, and companies do not pay much attention to it, resulting in a lack of active research.

To reduce this gap, we researched how to evaluate the charging process to improve the currently lacking EV charging experience (eCX). There's a widely recognized axiom in the business field: 'You can't manage what you don't measure'. Likewise, when it comes to managing eCX, defining the measurement elements and criteria is essential. So, this research aims to formulate effective heuristic evaluation criteria to assess eCX.

To gain an in-depth understanding of the EV charging process, hierarchical task analysis (HTA) was used. As a result, we found that eCX consists of multiple elements and interactions, including EVs, EV chargers, and mobile apps. This means we need different evaluation criteria for tasks and elements based on the time series, dynamically. However, existing heuristic evaluation methods focus on only one element and are evaluated with one heuristic (e.g., Nielsen's ten heuristics [19]). While Nielsen's heuristics are one of the dominant evaluation criteria, using a variety of heuristics

to evaluate new types of complex products could have identified more significant issues [3]. Therefore, we formulated evaluation criteria with various commonly used heuristics for each charging tasks and interaction elements necessary to assess the eCX.

Expert heuristic evaluations were conducted to verify if the developed evaluation criteria could effectively identify problems. Seven ergonomics professionals and stakeholders of EV charging service evaluated five EV chargers serving various charging methods. They assessed the chargers with the above criteria on a 7-point Likert scale and discussed eCX elements' UI and UX pros and cons. Then, the authors analyzed the rating data quantitatively and the discussions qualitatively concerning the statistical differences or not.

The analyses showed that the indexes with significant quantitative differences also had differences in the qualitative analysis, meaning selected heuristics can identify brand differences effectively. Many heuristic criteria have significant differences between brands in the Authentication phase but almost no difference in the Connection phase. Additionally, in the charging option selection phase, participants' mental models were observed to be similar to those of gas stations for internal combustion vehicles.

The significance and contribution of this research lies in evaluating the eCX and identifying effective human-computer interaction methods for the charging process through quantitative and qualitative analysis:

- We formulated dynamic evaluation criteria for EV charging tasks and components based on the time series. The eCX can be managed by assessed with these criteria to ensure well-designed EV charging systems.
- We observed pros and cons of several design aspects and user's mental model related to existing EV chargers.
- We could provide guidelines for the more user-friendly design of EV chargers, EV cars, and mobile applications to enhance the eCX.

2 RELATED WORKS

2.1 Usability of New Product or Service

When a new product category, such as smartphones and software applications, is introduced, previously established usability heuristics, such as Nielsen's [19], are often restructured to assess its usability. Examples include ambient displays [16], desktop applications [34], touchscreen-based mobile devices [10], mobile applications [8], and games [32]. These adaptations allow for the identification of more usability issues by modifying and supplementing various heuristics, all based on Nielsen's original set.

While all the examples mentioned above operate independently, a single product doesn't define the Electric Vehicle Charging Experience (eCX). If the connector doesn't smoothly fit into the EV's socket, is it an issue with the charger's usability or the car's? Similarly, if the charger's QR code isn't recognized correctly, is it a problem with its usability or the mobile app's? In other words, the eCX comprises three essential elements: EV charger, EV car, and mobile application.

Research on evaluating multi-component products or services, such as the Internet of Things (IoT) and Mobility as a Service (MaaS), is limited [13]. While Nielsen's heuristics are dominantly used for

usability evaluation, as mentioned earlier, adjusting heuristics to evaluate such complex services could help identify more significant issues.

2.2 Usability of Electric Vehicle

As EVs become more widespread, research on the user experience of EVs has also expanded. Most studies related to EVs have concentrated on optimizing infrastructure. This optimization involves tracking electricity demand over time, a crucial factor due to its significant correlation with prices [18]. Additionally, studies explore strategies to minimize operational costs by developing decentralized scheduling algorithms using deep learning [21]. Addressing such power grid challenges is essential for EVs to gain mainstream acceptance [14] and can also have an impact on user satisfaction [31]. Therefore, research on infrastructure remains a vital area significantly influencing usability.

Furthermore, there is research from a more user-centric perspective. 'Range anxiety' refers to concerns that EVs' limited battery capacity influences purchase decisions and usage patterns [22]. Similarly, 'gas anxiety' reflects the clear preference of plug-in hybrid vehicle drivers to minimize gasoline consumption [6]. Moreover, consumers who cannot install a private charging station at home may face a 'charging dilemma' when using public rapid chargers, often resulting in high charging fees [31]. However, these studies primarily focus on the usability of EVs, and there is limited discussion regarding the overall eCX.

3 METHOD

3.1 Hierarchical Task Analysis (HTA)

To systematically organize and evaluate the eCX, a Hierarchical Task Analysis (HTA) was performed [1, 4]. HTA is considered a structured exploration approach adaptable to various settings and objectives within the realm of human factors [26]. The eCX encompasses the EV car, EV charger, and mobile app. Unlike traditional products with relatively simple experiences, users of charging services navigate through complex task stages. To account for this complexity, tasks were categorized into three major levels. At the lowest level (Level 3), specific goals for each task were defined, and the devices involved in these interactions were identified.

3.2 Usability Index Selection

To determine the evaluation criteria for each stage, a matrix was employed to assess the relevance of heuristics as in Figure 1. Utilizing information processing stages [28] and established usability evaluation heuristics from existing literature, the heuristics were organized. Redundant or less relevant heuristics to EV charging were excluded, and the selected ones were arranged horizontally. Vertically, the stages of charging were listed, and each heuristic was rated based on its suitability for evaluation at each charging stage. The relevance to tasks at level 3 was assessed on a scale: 0 for 'not related at all,' 1 for 'related,' and 2 for 'highly related.' This evaluation involved three authors and two experts with a PhD in human factors engineering. Indices with an average score exceeding 1.5 were selected. Subsequently, elements considered obvious for the task (e.g., information availability in searching, etc.) were excluded to finalize the evaluation checklist.

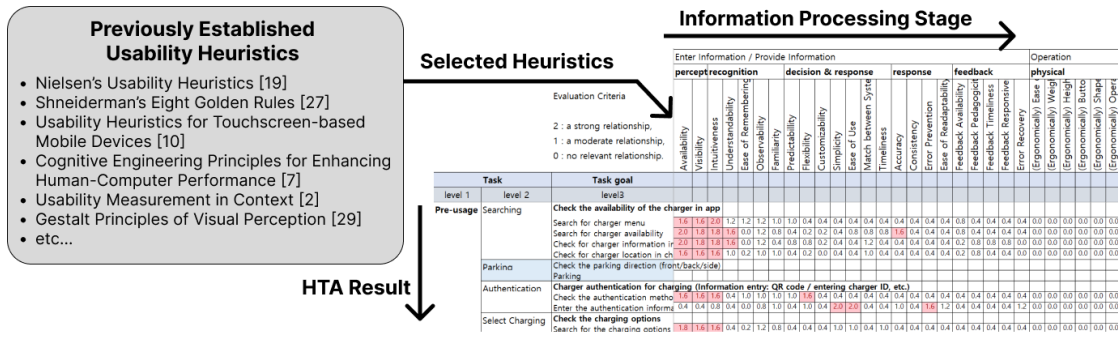


Figure 1: Process of Deriving Heuristics to Evaluate EV Charging. The first step involves selecting heuristics related to the EV charging process from the "Previously Established Usability Heuristics" [2, 7, 27, 29]. Using a matrix, these heuristics are horizontally listed according to the information processing stage, while the results of the Hierarchical Task Analysis (HTA) are listed vertically. This information is then used to assess the relevance and derive heuristics that need evaluation at each stage.

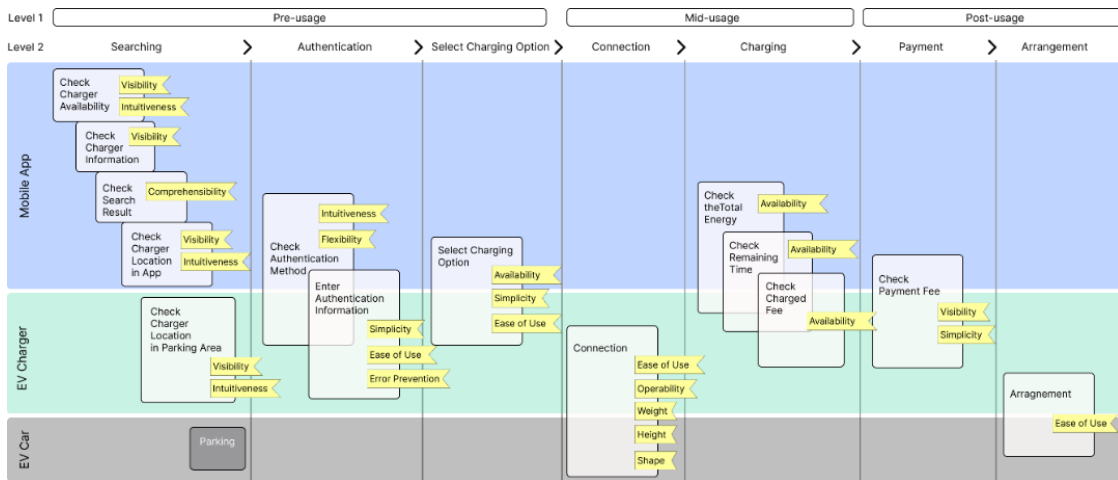


Figure 2: Results of HTA and Heuristic Index Selection. At the top are three level 1 tasks, along with seven level 2 tasks resulting from the HTA. The inner boxes represent the fourteen tasks corresponding to level 3, with heuristic indexes attached to their respective boxes.

3.3 Heuristic Evaluation

A heuristic evaluation was conducted to validate the appropriateness of the evaluation criteria. Our hypothesis posited that 'items with significant differences in UX will show statistically significant differences in evaluated ratings'. As detailed in Table 1 in the Appendix, participants included various experts in the human factors field, UI/UX designers employed at EV charging companies, and employees of automobile manufacturers. Teams comprised three to four individuals, and two sessions were conducted to involve as many stakeholders as possible.

The apparatuses used included the Tesla Model Y as an electric vehicle (EV) and the Galaxy S21 as a mobile phone. Five chargers were selected based on market share, characteristics, and charging methods. At the beginning of each session, the authors provided participants with detailed explanations of the scenarios and key points. Participants then proceeded to the nearest predetermined charging station, using the same five chargers, and experiencing

the entire charging process from the Searching phase to the Arrangement with the identical scenario. When chargers supported multiple authentication methods, the team tested all available methods.

The 'Think Aloud' technique was employed during charger use, with video recording in progress. After using each charger, debriefing sessions occurred in the car while traveling to the next site. Once all planned chargers were tested, the team returned to the lab to review recorded videos, photos, and clips, facilitating a detailed discussion. Throughout the discussion, the authors focused on evaluating the pros and cons of different methods of the charging process, variations between charger brands, and differences in charging locations.

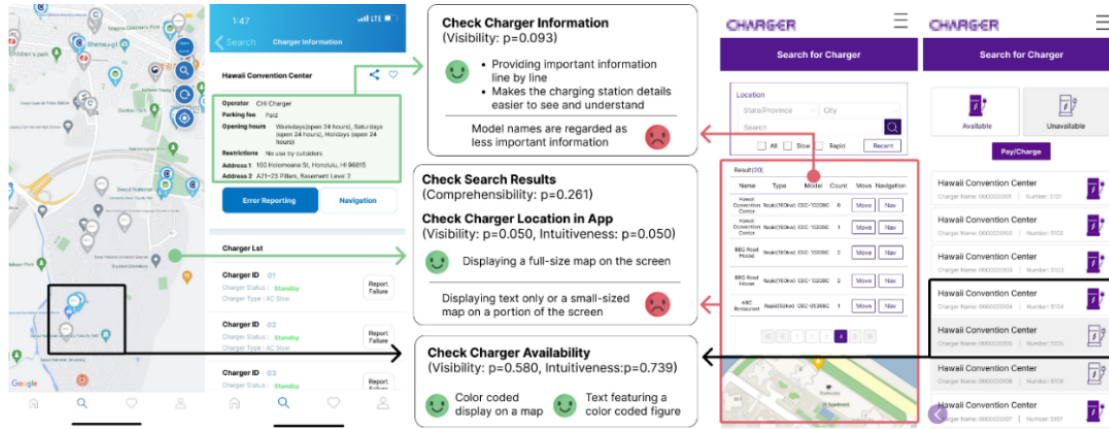


Figure 3: Searching Phase UI Comparison. In the Searching phase, the mobile app UIs of two brands were compared. The brand on the left received generally good heuristic evaluations, while the brand on the right received poor evaluations.

4 RESULT

4.1 HTA and Heuristic Index Extraction

Through HTA, we categorized the EV charging process into three hierarchical levels. At level 1, the highest classification, the charging process was broadly divided into four stages: First use, Pre-use, Mid-use, and Post-use. 'First use' pertains to the steps taken when using the charger for the first time, such as downloading the app, signing up, and registering a payment card. However, this stage was excluded as it occurs only once. At level 2, we further subdivided each level 1 category into seven sub-stages. For 'Pre-use,' these sub-stages include Searching, Authentication, and Select Charging Option Selection. 'Mid-use' includes Connection and Charging, and 'Post-use' includes Payment and Arrangement. At level 3, we specified the tasks that make up each stage in level 2. From this level, it is possible to determine through which medium each stage occurs. This is visually represented in Figure 2, where we organized the charging stages in columns and the means of user interaction in rows, placing level 3 tasks within this structure.

We derived 27 evaluation criteria for 14 tasks, enabling us to evaluate the usability of the eCX and identify issues along with tasks and components. These were constructed flexibly for each charging phase according to the characteristics of each eCX element.

4.2 Heuristic Evaluation

As a result of the heuristic evaluation, issues were identified at each stage. Since the evaluation involved seven participants and did not satisfy normality, a non-parametric test method, the Friedman chi-squared test, was conducted (See Table 2). This test aimed to verify differences between brands at each charging process stage using the selected heuristics.

4.2.1 Searching. In the 'Searching' phase, each step's heuristics and evaluation results are summarized below and illustrated in Figure 3.

- Check Charger Availability (Visibility: $\chi^2(4) = 2.871$, $p = 0.580$, Intuitiveness: $\chi^2(4) = 1.985$, $p = 0.739$): There is no

significant difference between indicating availability on the map using color coding and using text.

- Check Charger Information (Visibility: $\chi^2(4) = 7.966$, $p = 0.093$): The mobile app providing several significant information lines by line is better in visibility for charger information. If there were more non-important factors, the screen would become cluttered and lead to low visibility for important things.
- Check Search Results (Comprehensibility: $\chi^2(4) = 5.263$, $p = 0.261$): Displaying on the map is slightly better than displaying in text, but the difference is not statistically significant.
- Check Charger Location in the App (Visibility: $\chi^2(4) = 9.471$, $p = 0.050$, Intuitiveness: $\chi^2(4) = 9.477$, $p = 0.050$): Displaying the charger's location on the map can enhance visibility and intuitiveness, and it's better to display the map prominently.
- Check Charger Location in the Parking Area (Visibility: $\chi^2(4) = 4.098$, $p = 0.393$, Intuitiveness: $\chi^2(4) = 6.205$, $p = 0.184$): In the offline parking area, slightly better results are observed when chargers are marked distinctly, such as with color coding on the floor or pillars, but the difference is not statistically significant.

4.2.2 Authentication. In the 'Authentication' phase, each step's heuristics and evaluation results are summarized below and illustrated in Figure 4.

- Check Authentication Method (Intuitiveness: $\chi^2(4) = 18.496$, $p = 0.001^{***}$), Enter Authentication Information (Simplicity: $\chi^2(4) = 19.817$, $p = 0.001^{***}$, Ease of use: $\chi^2(4) = 20.0$, $p < 0.001^{***}$, Error prevention: $\chi^2(4) = 13.160$, $p = 0.011^*$): The left image in Figure 4 demonstrates a square shape for QR authentication, employing good affordance to make it intuitive, convenient, and error-preventive. In contrast, the images at the bottom in Figure 4 show the use of RFID card tagging, where poor affordance leads to a less intuitive, inconvenient, and error-prone experience for

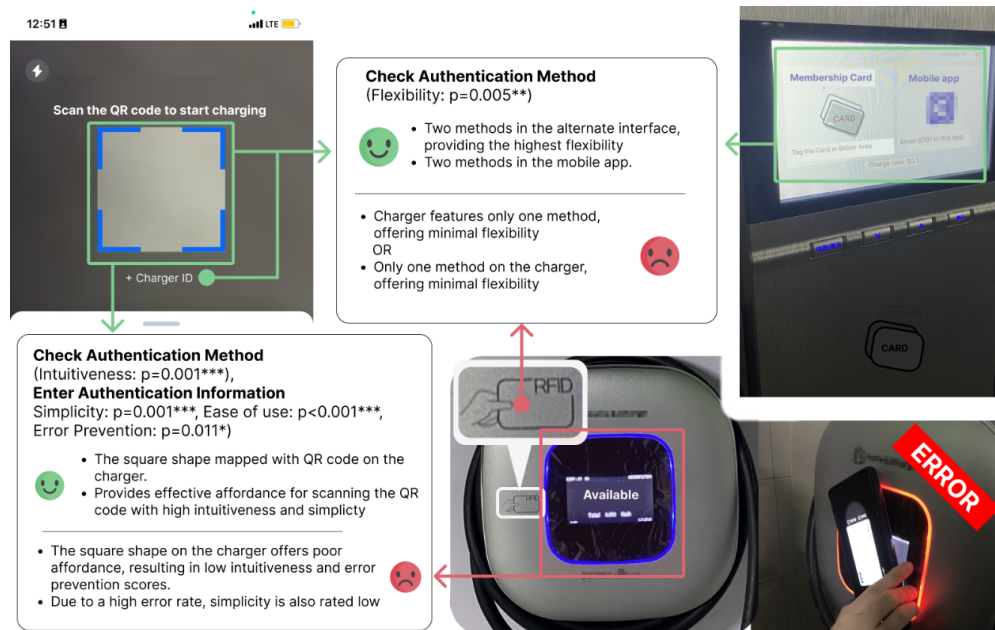


Figure 4: Authentication Phase UI/UX Comparison. In the Authentication phase, the mobile app and charger UI/UX of three brands were compared. The two brands above received generally good heuristic evaluations, while the lower brand received poor evaluation.

the user. Due to the high error rate, simplicity also scored low.

- Check Authentication Method (Flexibility: $\chi^2(4) = 14.885$, $p = 0.005^{**}$): Providing only one method for authentication was not effective. It was found that offering one method in each of two different mediums (e.g., one in the app – enter charger ID, and another in the charger – RFID card tagging) is more effective than providing two methods within a single medium.

4.2.3 Select Charging Option.

- Availability: There were three types: offering only price options (2 brands, as shown on the left in Figure 5 or like a gas station), multiple options provided (1 brand, as shown in the middle), and no options provided (2 brands, as shown in the right). In the case where only a price option is provided, some participants (2 and 3 people, respectively) responded that no options were provided.
- Simplicity ($\chi^2(4) = 6.0$, $p = 0.199$), Ease of use ($\chi^2(4) = 10.179$, $p = 0.038^{*}$): Cases where one can select options on the charger received high scores. In contrast, scores were low if provided in apps.

4.2.4 Connection. In the ‘Connection’ phase, each step’s heuristics and evaluation results are summarized below and illustrated in Figure 6.

- Connector (Ease of use: $\chi^2(4) = 5.426$, $p = 0.246$, Operability: $\chi^2(4) = 5.167$, $p = 0.271$, Weight: $\chi^2(4) = 2.807$, $p = 0.591$, Shape: $p=\chi^2(4) = 3.969$, $p = 0.410$): Regarding charging connectors, they are usually supplied by one manufacture

company, and most are from the same supplier. Therefore, there shouldn’t be significant differences in the heuristic scores, and indeed, there aren’t any major differences.

- Connector (Height: $\chi^2(4) = 13.061$, $p = 0.011^{*}$): However, the height of connectors can vary by charging company. If the height of the connector is ergonomically too low or too high, it receives poor scores.

4.2.5 Charging.

- Check Charged Energy (Availability): As in Figure 5, despite the available feature, the charged capacity is important information, but not easily discernible due to unnecessary large images in prominent positions.
- Check Remaining Time (Availability): Remaining charging time is not displayed if charging options are not selected.
- Check Charged Fee (Availability): It’s rare for the charged fee to be provided.

4.2.6 Payment.

- Check payment fee (Visibility: $\chi^2(4) = 5.684$, $p = 0.244$, Simplicity: $\chi^2(4) = 4.342$, $p = 0.362$): Although it was not statistically significant, the chargers that allowed immediate confirmation of payment amounts on the charger itself received good scores.

4.2.7 Arrangement.

- Ease of use ($\chi^2(4) = 2.092$, $p = 0.719$): Most slow chargers have light cables that are not very long, making them not difficult to arrange. Additionally, similar to the connection, there was not a significant difference between brands for the same reason.

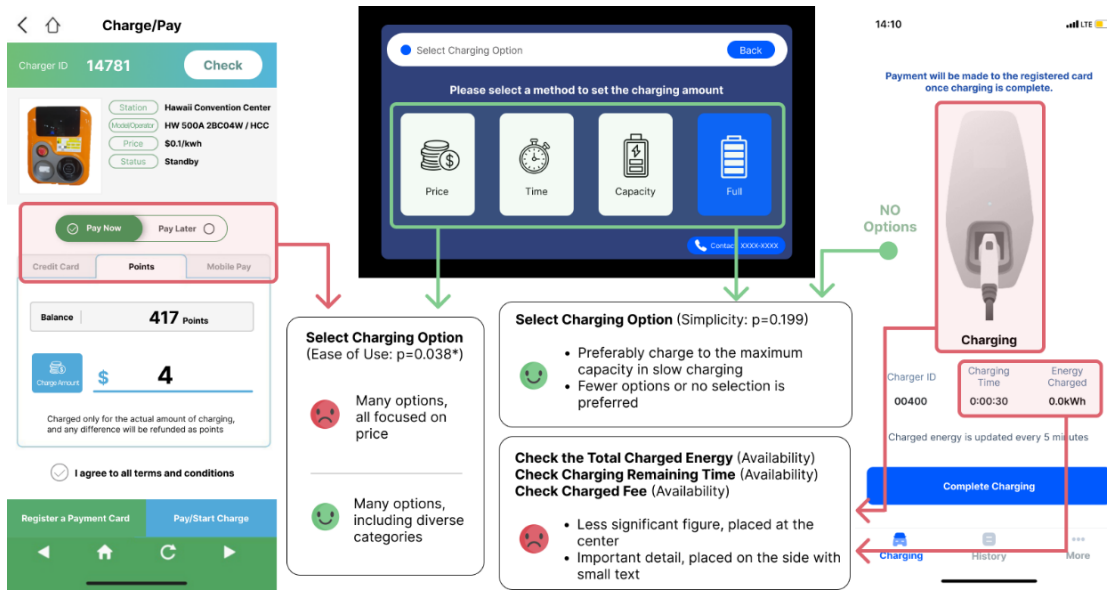


Figure 5: In the Select Charging Option and Charging phase, the UI of the mobile apps and chargers of three brands were compared. The middle brand received a generally good heuristic evaluation, while the brands on both sides received poor evaluation.

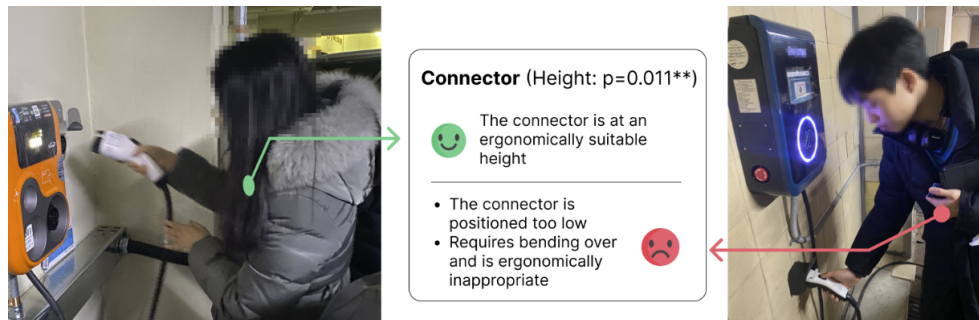


Figure 6: In the Connection stage, the UX of two chargers was compared. The brand on the left received generally good heuristic evaluations, while the brand on the right received poor evaluations.

5 DISCUSSION

This study aimed to enhance the Electric Vehicle Charging Experience (eCX) by identifying heuristic evaluation criteria to identify the usability issues of existing EV chargers for management. Since eCX is not confined to a single product, we restructured existing heuristics using a matrix. Seven experts then conducted heuristic evaluations to validate these heuristics. Using five different chargers, we evaluated based on derived metrics, enabling both quantitative and qualitative analysis of methods that significantly vary or not between brands.

Our research specifically evaluated eCX, a crucial aspect of EV usability. Previous studies have predominantly focused on EV usability, emphasizing charging power grids [18, 21] or addressing range anxiety [22]. However, charging is as important as the usability of the EV itself and is a critical factor in the proliferation of EVs [31]. Also, the eCX in this study involves complex tasks and

several components, organizing a kind of service. However, in traditional expert heuristic evaluations, a single metric was commonly used. To address this complexity, we applied a variety of heuristics and formulated criteria dynamically for each task and elements. Notably, we identified several issues in the current eCX, suggesting that some companies do not meet basic usability standards, indicating a significant oversight in UX aspects of EV charging. Inconsistent designs among charging companies have also troubled users [5, 12, 23]. This study provides guidelines for designing chargers, and companies can refer to these heuristics for designing or evaluating them.

An interesting observation is that participants still employ the mental model of a gas station when charging EVs. For chargers that offered price options only, some participants responded, 'There is no option available', while others indicated, 'The price option is available'. This response discrepancy is interpreted as participants

retaining the gas station's mental model (i.e., we only can select price) when charging EVs. Furthermore, selecting charging options via an app showed lower scores in 'Simplicity' and 'Ease of Use' than at the charger. This is attributed to the rarity of setting options and prices via an app at conventional gas stations, and participants are accustomed to choosing at the pump or counter. Therefore, if it becomes possible to select charging options on the EV charger screen based on price or full, similar to current gas stations, users could find it more convenient without conflicting with their existing mental model.

EV charging scenarios are highly diverse, encompassing both rapid and slow charging, with distinctions between charging at home, in an apartment, or at an office. Consequently, numerous chargers offer different eCX based on usage context [15, 25]. Conducting additional heuristics in these cases could yield valuable results following the User-Centered Design (UCD) methodology [17], enabling us to enhance eCX incrementally.

6 CONCLUSION

The proliferation of electric vehicles is expanding, and while numerous studies on EV user experiences are underway, there has been a notable absence of efforts to evaluate the usability of the EV charging process. Furthermore, the management of the Electric Vehicle Charging Experience (eCX) has often been overlooked, with priority placed on the development of EV technology. To address these challenges, we identified essential tasks and elements for managing slow EV chargers using HTA and formulated evaluation criteria. Following this, we conducted a heuristic evaluation to ensure these criteria were adequately extracted and could capture usability issues. Upon analyzing the results quantitatively and qualitatively, we found relatively good designs for eCX, and it will be even better if we follow the mental model of a gas station. This contribution aims to alleviate users' inconveniences due to inconsistent designs and assist companies in designing EV chargers and creating solutions.

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REFERENCES

- [1] Annett, J. Hierarchical task analysis. *Handbook of cognitive task design*, 2 (2003), 17-35.
- [2] Bevan, N. and Macleod, M. Usability measurement in context. *Behaviour & information technology*, 13, 1-2 (1994), 132-145.
- [3] Chalhoub, G., Kraemer, M. J. and Flechais, I. Useful shortcuts: Using design heuristics for consent and permission in smart home devices. *International Journal of Human-Computer Studies*, 182 (2024), 103177.
- [4] Diaper, D. and Stanton, N. *The handbook of task analysis for human-computer interaction* (2003).
- [5] Fabianek, P. and Madlener, R. Multi-Criteria assessment of the user experience at E-Vehicle charging stations in Germany. *Transportation Research Part D: Transport and Environment*, 121 (2023), 103782.
- [6] Ge, Y., MacKenzie, D. and Keith, D. R. Gas anxiety and the charging choices of plug-in hybrid electric vehicle drivers. *Transportation Research Part D: Transport and Environment*, 64 (2018), 111-121.
- [7] Gerhardt-Powals, J. Cognitive engineering principles for enhancing human-computer performance. *International Journal of Human-Computer Interaction*, 8, 2 (1996), 189-211.
- [8] Hecht, C., Figgenger, J. and Sauer, D. U. Analysis of electric vehicle charging station usage and profitability in Germany based on empirical data. *Iscience*, 25, 12 (2022).
- [9] IEA Global EV Outlook 2023. City, April, 2023.
- [10] Inostroza, R., Rusu, C., Roncagliolo, S., Jimenez, C. and Rusu, V. Usability heuristics for touchscreen-based mobile devices. *IEEE*, City, 2012.
- [11] Kano, N. Attractive quality and must-be quality. *Journal of the Japanese society for quality control*, 31, 4 (1984), 147-156.
- [12] Lee, M. L., Carter, S., Iliev, R., Bravo, N. S., Van, M. P., Denoue, L., Kimani, E., Filipowicz, A. L., Shamma, D. A. and Sieck, K. A. Understanding People's Perception and Usage of Plug-in Electric Hybrids. City, 2023.
- [13] Maas, B. Literature Review of Mobility as a Service. *Sustainability*, 14, 14 (2022), 8962.
- [14] Machineni, V. N., Vardhana, K. S. and Dhandapani, V. *Data Driven Approach to Achieve Coordinated Charging Among Electric Vehicles*. Springer, City, 2021.
- [15] Maguire, M. Context of use within usability activities. *International journal of human-computer studies*, 55, 4 (2001), 453-483.
- [16] Mankoff, J., Dey, A. K., Hsieh, G., Kientz, J., Lederer, S. and Ames, M. Heuristic evaluation of ambient displays. City, 2003.
- [17] Moumane, K., Idri, A. and Abran, A. Usability evaluation of mobile applications using ISO 9241 and ISO 25062 standards. *SpringerPlus*, 5 (2016), 1-15.
- [18] Needell, Z., Wei, W. and Trancik, J. E. Strategies for beneficial electric vehicle charging to reduce peak electricity demand and store solar energy. *Cell Reports Physical Science*, 4, 3 (2023).
- [19] Nielsen, J. *Enhancing the explanatory power of usability heuristics*. City, 1994.
- [20] Organization, W. M. July 2023 confirmed as hottest month on record. City, August 14, 2023.
- [21] Park, K. and Moon, I. Multi-agent deep reinforcement learning approach for EV charging scheduling in a smart grid. *Applied Energy*, 328 (2022), 120111.
- [22] Pevec, D., Babic, J., Carvalho, A., Ghiassi-Farrokhfal, Y., Ketter, W. and Podobnik, V. A survey-based assessment of how existing and potential electric vehicle owners perceive range anxiety. *Journal of Cleaner Production*, 276 (2020), 122779.
- [23] Recharge, S. *EV Driver Survey Report 2023*. 2023.
- [24] Sawaguchi, M. Innovation activities based on s-curve analysis and patterns of technical evolution—"From the standpoint of engineers, what is innovation?". *Procedia Engineering*, 9 (2011), 596-610.
- [25] Shackel, B. Usability—Context, framework, definition, design and evaluation. *Interacting with computers*, 21, 5-6 (2009), 339-346.
- [26] Shepherd, A. HTA as a framework for task analysis. *Ergonomics*, 41, 11 (1998), 1537-1552.
- [27] Shneiderman, B. Designing for Fun: Can we design user interfaces to be more fun?(2004). 2005.
- [28] Simon, H. A. Information processing models of cognition. *Annual review of psychology*, 30, 1 (1979), 363-396.
- [29] Todorovic, D. Gestalt principles. *Scholarpedia*, 3, 12 (2008), 5345.
- [30] Venkitesh, A. K. and Kosuru, V. S. R. A review on autonomous electric vehicle communication networks-progress, methods and challenges. *World Journal of Advanced Research and Reviews*, 16, 3 (2022), 013-024.
- [31] Wang, Y., Chi, Y., Xu, J.-H. and Yuan, Y. Consumers' attitudes and their effects on electric vehicle sales and charging infrastructure construction: an empirical study in China. *Energy Policy*, 165 (2022), 112983.
- [32] Yanez-Gomez, R., Font, J. L., Cascado-Caballero, D. and Sevillano, J.-L. Heuristic usability evaluation on games: a modular approach. *Multimedia Tools and Applications*, 78 (2019), 4937-4964.
- [33] Yoon, G., Kim, S., Shin, H., Cho, K., Jang, H., Lee, T., Choi, M.-i., Kang, B., Park, S. and Lee, S. Carbon-Neutral ESG Method Based on PV Energy Generation Prediction Model in Buildings for EV Charging Platform. *Buildings*, 13, 8 (2023), 2098.
- [34] Zhang, D. and Adipat, B. Challenges, methodologies, and issues in the usability testing of mobile applications. *International journal of human-computer interaction*, 18, 3 (2005), 293-308.

A APPENDICES

A PROFILES OF EXPERT HEURISTIC EVALUATION PARTICIPANTS

Table 1: Profiles of the experts in each field who participated in the heuristic evaluations

Participant no.	Position	Experience (years)	Session
P1	EV Charger Company UX Designer	4	A
P2	PhD in Human Factor, University Lecturer	10	
P3	MS in Human Factor, UX Researcher	5	
P4	Major in Human Factor	2	
P5	EV Charger Company UI Designer	5	B
P6	MS in Human Factor Expert, Vehicle Company UX Designer	7	
P7	Major in Human Factor	3	

B THE ECX EVALUATION CRITERIA AND RESULT OF QUANTITATIVE ANALYSIS

Table 2: The formulated eCX evaluation heuristics and result of quantitative analysis of expert heuristic evaluation.

Task Level 2	Task Level 3	Heuristics	Statistics	p-value
Searching	Check Charger Availability	Visibility	2.871	0.580
		Intuitiveness	1.985	0.739
	Charger Information	Visibility	7.966	0.093
	Check Search Results	Comprehensibility	5.263	0.261
	Check Charger Location in App	Visibility	9.471	0.050
		Intuitiveness	9.477	0.050
	Check Charger Location in Parking Area	Visibility	4.098	0.393
		Intuitiveness	6.205	0.184
Authentication	Check Authentication Method	Intuitiveness	18.496	0.001
		Flexibility	14.885	0.005
	Enter Authentication Information	Simplicity	19.817	0.001
		Ease of use	20.000	<0.001
		Error prevention	13.160	0.011
Select Charging Option	Availability	-	-	
	Simplicity	Simplicity	6.000	0.199
	Ease of use	Ease of use	10.179	0.038
Connection	Connector	Ease of use	5.426	0.246
		Operability	5.167	0.271
		Weight	2.807	0.591
		Height	13.061	0.011
		Shape	3.969	0.410
Charging	Check the Total Charged Energy	Availability	-	-
	Check Charging Remaining Time	Availability	-	-
	Check Charged Fee	Availability	-	-
Payment	Check Payment Fee	Visibility	5.684	0.224
		Simplicity	4.342	0.362
Arrangement	Ease of use	Ease of use	2.092	0.719